Verónica Castañeda

PhD Candidate in Biomedicine | Universidad de los Andes, Chile

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Molecular biologist and bioinformatician driven by curiosity, scientific rigor, and the conviction that complex problems can be solved through interdisciplinary collaboration. My research experience spans stem cell biology, regenerative medicine, transcriptomics, and space biology, combining experimental and computational approaches across Ecuador, Chile, and the United States. As a PhD candidate in Biomedicine and active member of NASA GeneLab collaborations, I have contributed to a growing body of scientific publications while developing a strong commitment to teaching, mentorship, and scientific communication. My long-term goal is to establish an internationally connected research program in Ecuador that leverages bioinformatics to advance regenerative medicine, healthy aging, and space life sciences.

Highlights

- PhD Candidate in Biomedicine (expected graduation September 2026)
- 14 peer-reviewed publications
- Springer Nature Article of the Year Award (2025)
- NASA GeneLab Multiomics Analysis Working Group member
- Recipient of PhD fellowships from ANID, IMPACT, and UANDES
- Recipient of training grants from NASA OSDR AWG and EMBO
- Research experience in Ecuador, Chile, and the United States
- Teaching and mentoring experience in molecular biology and bioinformatics

EDUCATION

PhD Candidate, Universidad de los Andes, Chile **2021- 2026**

- Doctoral thesis at the Laboratorio de Biología Molecular y Bioinformática: “Identification of key lncRNA for chondrocyte differentiation” with Gino Nardocci, PhD
- Doctoral internship Albert Einstein College of Medicine: “Fate commitment of murine BM-MSCs to chondrogenic phenotype” with Maria Maryanovich, PhD
- Rotation at Laboratorio de Medicina Regenerativa: “Effect of mitochondrial transplant on oxidative stress in osteoarthritis” with Maroun Khoury, PhD

Engineer in Biotechnological Processes, Universidad San Francisco de Quito, Ecuador **2015- 2020**

- Undergrad thesis: "POU1F1 mutations and hypopituitarism: Relationship with the clinical phenotype of hypopituitarism, molecular modeling and functional prediction” with Andrés Caicedo, PhD
- Research volunteer in Ecuadorian patrimonial plants initiatives under the mentorship of Carlos Ruales.
- Internship with Dr. Sonia Zapata at the Microbiology Institute
- Honors program, GPA 3.53/4.00 Cum laude.

TEACHING EXPERIENCE

Universidad de los Andes, Chile: Teaching Assistant – Molecular and Cellular Biology **2023–2025**

- Developed and evaluated assessments, facilitated discussion-based learning sessions, and mentored doctoral students in molecular and cellular biology.
- Provided academic support by leading study group sessions for PhD students.
- Direct guided discussion of scientific articles to further the student’s understanding.

ID Core Biotechnology, Ecuador: Bioinformatics teacher **2020–2021**

- Preparing tests, homework, academic material, and grading
- Taught an international audience online about protein modeling

Universidad San Francisco de Quito, Ecuador: **2017-2019**

- Teacher assistant in Botany exercises class: practical exercises, student assessment, and academic mentoring.
- Participation as tutor for the Learning center
- Mentoring students with Andrés Caicedo online with bioinformatic analysis and in the laboratory

RESEARCH EXPERIENCE

Active member at NASA's GeneLab, USA- Washington DC **2022 - Present**

Selected achievements

- NASA Open Science 101 Certificate of Achievement (2024)
- Co-author on international collaborative publications in aging and space biology
- Transcriptomic analysis of sarcopenia and frailty datasets

Instituto de Investigación en Biomedicina, Quito - Ecuador **2018-2021**

Thesis student and Undergraduate research assistant

- Worked under the guidance of Dr. Andrés Caicedo
- Acquired cell culture experience and by performing experiments and analyzing results which my first peer reviewed article.
- Led my research team where we compiled clinical reports and used protein structure modeling to study missense mutations.

Keck Graduate Institute, USA -California **Summer 2019**

Summer intern

- Worked with the guidance of Dr. Ilya Tolstorukov
- Improved S. cerevisiae knock out strategy during the Summer Undergraduate Research Experience program

SELECTED PUBLICATIONS (*first or co-first)

1. Kevin Zambrano … **Verónica Castañeda**… Andrés Caicedo (2026). Anesthesia and mitochondria: balancing toxicity and protection through emerging therapeutic strategies**.** J Anesth.

2. Lourdes M. Orejuela-Escobar, Carolina Tituaña, Arleth Gualle, Andrea C. Landázuri, Edgar J. Guerrón-Orejuela, Emilia Morales, **Verónica Castañeda**, Andrés Caicedo (2026). Colored avocado (*Persea americana* Mill.) seed extract using a natural deep eutectic solvent for potential sustainable food coloring applications: towards a circular bioeconomy. RSC.

3. Aliosha Figueroa-Valdés,… **Verónica Castañeda**,…Maroun Khory & Francisca Alcayaga (2025). Clinical-grade extracellular vesicles derived from umbilical cord mesenchymal stromal cells: preclinical development and first-in-human intra-articular validation as therapeutics for knee osteoarthritis. J Nanobiotechnology.

4. Christopher E. Mason, … **Veronica Castañeda**, …Afshin Beheshti (2024). A second space age spanning omics, platforms and medicine across orbits. Nature.

5. Andrea Camera, Marshall Tabetah, **Veronica Castañeda (*)**, …Afshin Beheshti (2024). Aging and putative frailty biomarkers are altered by spaceflight. Scientific Reports.

6. Francisco Cabrera, **Verónica Castañeda**, ... & Caicedo, A. (2022). Early evidence of the artificial transfer/transplant of mitochondria to oocytes and zygotes by MitoCeption. Mitochondrion.

7. Ariana León-Sosa, **Verónica Castañeda (*)**, … Caicedo, A. (2022). Key points for translating wound regenerative agents from in vivo assays in mice to clinical validation. Cytotherapy.

8. **Verónica Castañeda (*)**., Haro-Vinueza, A., Salinas, I., Caicedo, A., & Méndez, M. Á. (2022). The MitoAging Project: Single nucleotide polymorphisms (SNPs) in mitochondrial genes and their association to longevity. Mitochondrion.

9. Andrea Landázuri,… **Verónica Castañeda,** … Andrés Caicedo, & Lourdes Orejuela-Escobar, (2021). Moringa oleifera Lam. leaf powder antioxidant activity and cytotoxicity in human primary fibroblasts. Natural Product Research.

10. Francesca Velarde, **Verónica Castañeda ,** … & Caicedo, A. (2020). Use of human umbilical cord and its byproducts in tissue regeneration. Frontiers in bioengineering and biotechnology.

ABSTRACTS PRESENTED IN CONFERENCES

- ISCT, New Orleans Symposium (May, 2021)
- Mitochondrial DNA SNP's promoting longevity: Systematic Literature Review
- EMBL The Complex Life of RNA (October, 2022)
- Poster: Identification of novel lncRNAs, EMBO travel grant
- BSCP Harvard Medical School, NESS (April, 2022)
- Poster: Mitochondrial DNA missense mutations
- NYSCF conference (October, 2023) Poster: Harnessing lncRNAs in Chondrogenesis, UANDES travel grant
- Chilean Society for Biochemistry and Molecular Biology, La Serena (October, 2023) Poster: Exploring the Regulatory Role of lncRNAs in Chondrogenesis: Towards Targeted Mesenchymal Stem Cell Phenotype Modulation. UANDES travel grant

AWARDS

- Siemens Engineers for Change, E4C prize (September, 2020): Collaborated with mechanic and civil engineers to design a sustainable desalination project. Desalination program for Santa Elena, Esmeraldas, Ecuador
- BRAIN Chile winners (September, 2020): Wound healing therapy, foundation of Dragon BioMed
- Academic Excellence Award (2021) UANDES PhD Program in Biomedicine
- Springer Nature (2025) Article of the Year Award Clinical-grade extracellular vesicles derived from umbilical cord mesenchymal stromal cells: preclinical development and first-in-human intra-articular validation as therapeutics for knee osteoarthritis.

SKILLS

- Molecular and Cellular Biology: RT-qPCR, Mammalian Cell Culture, Knockdown, Overexpression, Western Blot
- Bioinformatics and Data Science: R Programming (UANDES 2025), Machine Learning (UANDES 2025), Python Programming, Transcriptomics, Multi-omics Data Analysis, Statistical Analysis
- Languages: Spanish (Native), English (C2), French (B2), German (A2)

VOLUNTEER WORK

- Out-reach efforts: To improve digital communication I was editor and writer of the "Revista CiiB". Additionally, I created a webpage for my laboratory to increase the visibility of our work. Finally, I prepared and oriented scientific workshops with high schoolers in collaboration with my Principal Investigator.
- TuTutor: Taught mathematics and English to children affected by the COVID-19 pandemic who were unable to attend classes

REFERENCES

Gino Nardocci, PhD
Professor and Researcher,
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Dolores Busso, PhD
Professor, Researcher and Director of PhD in Biomedicine Program
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Andrés Caicedo, PhD
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